

Maharashtra Public Service Commission Assessment

Subject Name :	Assistant Professor Zoology Maharashtra Education Service Gr A Collegiate Branch
Duration :	60
Number of Questions :	100
Total Marks :	200
Display Marks:	Yes
Maximum Instruction Time :	0
Is Section Default? :	No

Question Number : 1 Question Id : 464662801 Question Type : MCQ

Match the following accessory respiratory organs of vertebrates with their representative example.

- | | |
|--------------------------|----------------|
| a. Skin | i. Lepidosiren |
| b. Cloacal bladder | ii. Salamander |
| c. Branchial diverticula | iii. Turtle |
| d. Pelvic gills | iv. Clarias |

Options :

1. a-iv b-ii c-i d-iii
2. a-iii b-i c-ii d-iv
3. a-i b-iv c-iii d-ii
4. a-ii b-iii c-iv d-i

Match the following A with B group.

A

- a. Latitudinal migration
- b. Longitudinal migration
- c. Altitudinal migration
- d. Partial migration

B

- i. Migration of birds from East to West and vice versa
- ii. Migration of birds in summer but not in winter
- iii. Migration of birds from North to South and vice versa
- iv. Migration of birds from high mountain to low valley

Options :

- 1. a-iv b-ii c-iii d-i
- 2. a-ii b-iii c-i d-iv
- 3. a-iii b-i c-iv d-ii
- 4. a-i b-iv c-ii d-iii

Question Number : 3 Question Id : 464662803 Question Type : MCQ

In *Ascaris lumbricoides* seminal vesicle functions as _____.

Options :

1. Storage of sperm
2. Fertilization
3. Protection
4. None of these

Question Number : 4 Question Id : 464662804 Question Type : MCQ

Normal life span of *Ascaris* is _____.

Options :

1. 9 – 12 Weeks
2. 9 – 12 Months
3. 9 – 12 Years
4. 1 – 2 Months

Question Number : 5 Question Id : 464662805 Question Type : MCQ

Liver fluck (fasciola hepatica) belongs family _____.

Options :

1. Schistosomatidae
2. Bucephalidae
3. Fasciolidae
4. None of these

Question Number : 6 Question Id : 464662806 Question Type : MCQ

The life cycle of monocystis is monogenetic and characterized by the absence of sexual reproduction by _____.

- a. Gametogony
- b. Syngamy
- c. Sporogony
- d. Schizogony

Options :

- 1. a only
- 2. a and b only
- 3. a, b and c only
- 4. d only

Question Number : 7 Question Id : 464662807 Question Type : MCQ

Obelia shows polymorphism in which polyps are used for _____.

Options :

1. Budding
2. Feeding
3. Disseminating gametes
4. Defense

Question Number : 8 Question Id : 464662808 Question Type : MCQ

In honey bee nuptial flight occurs for the _____.

Options :

1. Foraging
2. Swarming
3. Mating
4. Defense

Question Number : 9 Question Id : 464662809 Question Type : MCQ

Which is the larva of Leucosolenia ?

Options :

1. Parenchymula
2. Amphiblastula
3. Stomoblastula
4. Coeloblastula

Question Number : 10 Question Id : 464662810 Question Type : MCQ

Newly hatched larva of *Ascaris lumbricoides* measures about _____.

Options :

1. 0.1 to 0.2 mm
2. 0.2 to 0.3 mm
3. 0.3 to 0.4 mm
4. None of these

Question Number : 11 Question Id : 464662811 Question Type : MCQ

In male mosquitoes aedeagus project posteriorly formed by fusion of _____.

- a. Gonapophyses of 10th segment
- b. Ciri of 10th segment
- c. Gonapophyses of 9th segment
- d. Claspers of 9th segment

Options :

- 1. a, b and d only
- 2. b, c and d only
- 3. a only
- 4. c only

Question Number : 12 Question Id : 464662812 Question Type : MCQ

Movement in pila is produced by _____.

- a. Blood flow only
- b. Waves of muscular contraction
- c. Ciliary action
- d. Flagellar movement

Which is the statement given above is/are correct ?

Options :

- 1. a, b and d
- 2. b only
- 3. c only
- 4. b, c and d

Question Number : 13 Question Id : 464662813 Question Type : MCQ

Brachiolaria larva is found in the life cycle of which animal ?

- a. Asterias
- b. Leech
- c. Unio
- d. Prawn

Options :

1. a, b and c only
2. b, c and d only
3. c and d only
4. a only

Question Number : 14 Question Id : 464662814 Question Type : MCQ

Placoid scales are the characteristic of _____.

Options :

1. Eel fish
2. Shark fish
3. Trygon fish
4. None of these

Question Number : 15 Question Id : 464662815 Question Type : MCQ

Herdmania is _____ animal.

Options :

1. Unisexual
2. Hermaphrodite
3. Asexual
4. None of these

Question Number : 16 Question Id : 464662816 Question Type : MCQ

Consider the following statements :

- a. Paedomorphosis is the retention of those characters which occurred in younger stages of the ancestors of a sexually mature descendant.
- b. Neoteny is due to a low level of Thyrotropin Releasing Hormone (TRH).

Options :

- 1. Neither of the statement is correct
- 2. The first statement is correct not the second
- 3. Both the statements are correct
- 4. The second statement is correct not the first

Question Number : 17 Question Id : 464662817 Question Type : MCQ

Reptiles flourished in the _____ era with gaint dinosaurs, dominating earth and seas.

Options :

1. Paleozoic era
2. Mesozoic era
3. Cenozoic era
4. None of these

Question Number : 18 Question Id : 464662818 Question Type : MCQ

Match the anatomical/physiological feature corresponding to flight adaptation in birds.

- | | |
|--------------------|---|
| a. Air sacs | i. Light weight, hollow structure |
| b. Pneumatic bones | ii. Reduced weight |
| c. Keeled sternum | iii. Efficient respiration to liberate large amount of energy |
| d. Single ovary | iv. Attachment of large flight muscles |

Options :

1. a-i b-ii c-iii d-iv
2. a-iii b-i c-iv d-ii
3. a-ii b-iv c-i d-iii
4. a-iv b-iii c-ii d-i

Question Number : 19 Question Id : 464662819 Question Type : MCQ

_____ is small arboreal, nocturnal and insectivorous marsupial found in Australia, North and South America.

Options :

1. Shrew
2. Hedgehog
3. Opossum
4. Kangaroo

Question Number : 20 Question Id : 464662820 Question Type : MCQ

_____ vertebra has its centrum concave at both the ends.

Options :

1. Amphicoelous
2. Procoelous
3. Opisthocoelous
4. Heterocoelous

Question Number : 21 Question Id : 464662821 Question Type : MCQ

The body surface of sponges is perforated by _____ through which water enters the body.

Options :

1. Oscullam
2. Ostia
3. Spongocoel
4. None of the above

Question Number : 22 Question Id : 464662822 Question Type : MCQ

During respiration in cockroaches the muscles responsible for rhythmic contraction and expansion of abdomen is _____.

- a. Tergo-sternal muscles
- b. Tergo muscle
- c. Sternal muscle
- d. Contract muscle

Which of the statement given above is/are correct ?

Options :

- 1. a only
- 2. b, c and d
- 3. a and c only
- 4. d only

Question Number : 23 Question Id : 464662823 Question Type : MCQ

Number of pairs of gill-pouches in scoliodon _____.

- a. 4
- b. 5
- c. 6
- d. 7

Which of the statement given above is/are correct ?

Options :

- 1. a only
- 2. a, c and d
- 3. b only
- 4. a and d only

Question Number : 24 Question Id : 464662824 Question Type : MCQ

Wuchereria bancrofti belongs class _____.

Options :

1. Phasmidia
2. Cestoda
3. Nematoda
4. None of these

Question Number : 25 Question Id : 464662825 Question Type : MCQ

In case of cnidaria, the small fresh water polyp is called as

Options :

1. Leech
2. Hydra
3. Starfish
4. None of these

Question Number : 26 Question Id : 464662826 Question Type : MCQ

The Hardy-Weinberg law is a basic concept in the population genetics.

- a. Hardy-Weinberg law developed independently in 1905.
- b. The law state that gene frequencies and genotypic ratio in natural breeding population remain constant from generation to generation.
- c. Hardy-Weinberg law is used to determine number of harmful mutation in population is increasing.
- d. In nature the Hardy-Weinberg law is impossible.

Which of the statement given above is/are correct ?

Options :

- 1. a only
- 2. b, c and d
- 3. b only
- 4. a, b and c

Question Number : 27 Question Id : 464662827 Question Type : MCQ

Which of the following interaction is correctly matched with its effects on the interacting species ?

Options :

1. Mutualism (+ -)
2. Commensalism (+ 0)
3. Competition (+ +)
4. Predation (+ +)

Question Number : 28 Question Id : 464662828 Question Type : MCQ

Fluorescence in situ hybridization (FISH) is used for _____.

Options :

1. RNA translation
2. Protein detection
3. DNA sequence localization on chromosomes
4. Enzyme Kinetics

Question Number : 29 Question Id : 464662829 Question Type : MCQ

Arrange the following in the correct evolutionary sequence.

- a. Homo habilis
- b. Australopithecus africanus
- c. Homo erectus
- d. Homo sapiens

Options :

- 1. $b \rightarrow a \rightarrow c \rightarrow d$
- 2. $a \rightarrow b \rightarrow c \rightarrow d$
- 3. $b \rightarrow c \rightarrow a \rightarrow d$
- 4. $c \rightarrow b \rightarrow a \rightarrow d$

Question Number : 30 Question Id : 464662830 Question Type : MCQ

The exciting potential of technique lies in the possibility of correcting genetics defects for example restoring the ability of a diabetic individual to make insulin or correcting genetics defects such technique is called as _____.

Options :

1. Gene therapy
2. Genetic counselling
3. Sterilization
4. None of these

Question Number : 31 Question Id : 464662831 Question Type : MCQ

The definition of species as groups of inter breeding natural populations that is reproductively isolated from other such groups is known as the _____.

Options :

1. Phylogenetic Species Concept
2. Evolutionary Species Concept
3. Recognition Species Concept
4. Biological Species Concept

Question Number : 32 Question Id : 464662832 Question Type : MCQ

One of patterns that do not confirm the historical reality of evolution and that make sense only if evolution has occurred is _____.

Options :

1. Homology
2. Vestigial structures
3. Fossil record
4. Punctuated equilibrium

Question Number : 33 Question Id : 464662833 Question Type : MCQ

In India the Nilgiri biosphere reserve include which of the following parts of State ?

Options :

1. Karnataka
2. Kerala
3. Tamil Nadu
4. All of these

Question Number : 34 Question Id : 464662834 Question Type : MCQ

Hardy Weinberg law depends on which of the following genetic equilibriums for its full attainment ?

Options :

1. No selection is operative
2. Random mating
3. Population is closed
4. All of these

Question Number : 35 Question Id : 464662835 Question Type : MCQ

Benthic region of marine ecosystem. It is the sea bottom from the sea shore to the deepest part of the sea.

- a. Littoral Zone → Abyssal Zone → Hadal Zone → Bathyl Zone
- b. Bathyl Zone → Littoral Zone → Abyssal Zone → Hadal Zone
- c. Littoral Zone → Bathyl Zone → Abyssal Zone → Hadal Zone
- d. Hadal Zone → Abyssal Zone → Littoral Zone → Bathyl Zone

Which of the statement given above is/are correct ?

Options :

- 1. a only
- 2. c only
- 3. b only
- 4. d only

Question Number : 36 Question Id : 464662836 Question Type : MCQ

When random genetic drift might trigger the origin of new species then it is called as _____.

Options :

1. Peripatric speciation
2. Speciation by genetic conflict
3. Mutation order speciation
4. Speciation by polyploidy

Question Number : 37 Question Id : 464662837 Question Type : MCQ

Transgenic Salmon is a genetically modified organism

- a. Transgenic Salmon have been created that overproduce growth hormone leading to extraordinary growth rate.
- b. AquaBounty Technologies from Waltham Massachusetts developed transgenic Salmon.
- c. In September 2013 US Food and Drug Administration (FDA) accepted Transgenic Salmon as a safe food.
- d. Transgenic Salmon developed from hybrid Atlantic Salmon and Chinook Salmon.

Which of the statement given above is/are correct ?

Options :

- 1. a only
- 2. b only
- 3. a, b and c
- 4. a, b and d

Question Number : 38 Question Id : 464662838 Question Type : MCQ

The population growth can be described by the following logistic equation

$$\frac{dN}{dt} = rN \left(\frac{K - N}{K} \right) \text{ where 'K' represent } \underline{\hspace{2cm}}.$$

Options :

1. Mortality
2. Natality
3. Carrying capacity
4. None of the above

Question Number : 39 Question Id : 464662839 Question Type : MCQ

The large biogeochemical areas having distinct climate, vegetation and animals are called biomes.

- a. The various zoogeographic regions contain more than one biome.
- b. There are two biomes terrestrial and aquatic.
- c. The major biomes are grass land, desert, arctic, mountain and forest biome.
- d. Nearly 10% of the earth is covered by desert.

Which of the statement given above is/are correct ?

Options :

- 1. a only
- 2. a, b and c
- 3. a and d
- 4. a, b and d

Aquatic ecosystem is largest ecosystem in the world.

- a. Water covers about $\frac{3}{4}^{\text{th}}$ of the earth surface.
- b. The ocean water is salty. It contain about 45 gm of salt in 1000 ml of water.
- c. The ecosystem of sea water is called marine ecosystem.
- d. The maximum depth of sea water is 8924 m.

Which of the statement given above is/are correct ?

Options :

- 1. a only
- 2. a and c
- 3. b and c
- 4. a, b and d

Question Number : 41 Question Id : 464662841 Question Type : MCQ

The pearl oysters are bivalves with two valves of the shell.

- a. They belongs to phylum Mollusca.
- b. Pinctada Fucata is the most dominant species in Indian waters.
- c. Pearl-Oysters are not a filter feeders.
- d. Pearl is the only gem produced by a living animal.

Which of the statement given above is/are correct ?

Options :

- 1. a only
- 2. b only
- 3. a, b and c
- 4. a, b and d

Question Number : 42 Question Id : 464662842 Question Type : MCQ

Lac is a natural resin produced by the lac insect.

- a. Indian lac insect Tachardia lacca belongs to order Diptera.
- b. Nail Polish is a good example of by product of Lac.
- c. Shellac is used for varnishes paints and polishes.
- d. Shellac is used in electrical industry as insulating material.

Which of the statement given above is/are correct ?

Options :

- 1. a only
- 2. b only
- 3. a, b and d
- 4. b, c and d

Question Number : 43 Question Id : 464662843 Question Type : MCQ

The chemical energy stored as nutrients in the bodies and wastes of organisms flows through eco system from one trophic (feeding) level to another. In this regard oriental magpie Robin falls in _____.

Options :

1. detritus feeders
2. second trophic level
3. third trophic level
4. fourth trophic level

Question Number : 44 Question Id : 464662844 Question Type : MCQ

The honey produced by bees from the nectar of Rhododendron is named as _____.

Options :

1. honey dew honey
2. blended honey
3. toxic honey
4. ideal honey

Question Number : 45 Question Id : 464662845 Question Type : MCQ

Which human is known as “hardy man” ?

Options :

1. Homohabilis
2. Homosapiens
3. Homoerectus
4. Australopithecus

Question Number : 46 Question Id : 464662846 Question Type : MCQ

The HIV virus critically injures the immune system by infecting and killing which cells ?

Options :

1. Platelets
2. RBCs
3. Alpha cells
4. T-cells

Question Number : 47 Question Id : 464662847 Question Type : MCQ

Which of the following is cat-fish ?

Options :

1. Labeo
2. Wallago attu
3. Mrigala
4. Catla

Question Number : 48 Question Id : 464662848 Question Type : MCQ

A cat fish species in a pond, sharing the pond with other fish species and with plants, insects, ducks and many other species makes up the _____.

Options :

1. eco system
2. population
3. biosphere
4. biological community

Question Number : 49 Question Id : 464662849 Question Type : MCQ

The following factors are influencing population dynamics.

- a. Natality
- b. Mortality
- c. Ecological niche
- d. Population density

Which of the statement given above is/are correct ?

Options :

- 1. a only
- 2. b only
- 3. a, b and c
- 4. a, b and d

Question Number : 50 Question Id : 464662850 Question Type : MCQ

In rivers the fishes commonly occurring are called carps.

- a. Rohu, Catla, Mrigal are common fishes in rivers.
- b. The major carps grow to a large size from 1 meter to 2m in length.
- c. The major carps mainly herbivorous feeds on algae and plants.
- d. Freshwater fisheries can be broadly divided into Esturine fisheries and riverine fisheries.

Which of the statement given above is/are correct ?

Options :

- 1. a only
- 2. b only
- 3. a, b and c
- 4. a, c and d

Question Number : 51 Question Id : 464662851 Question Type : MCQ

Which of the following is an inhibitor of voltage-gated sodium (Na^+) channels ?

Options :

1. Tetrodotoxin
2. Digitoxin
3. α -Bungarotoxin
4. Batrachotoxin

Question Number : 52 Question Id : 464662852 Question Type : MCQ

Which of the following proteins-DNA binding motif combination is incorrect ?

Options :

1. TFIIIA-Zinc Finger
2. Myo D-Basic helix-loop-helix
3. Jun-Leucine zipper
4. Cro-Zinc Finger

Question Number : 53 Question Id : 464662853 Question Type : MCQ

DNA polymerase selects the correct nucleotide primarily based on _____.

- a. Recognition of nucleotide sequence
- b. Shape of the base pair formed
- c. Energy of phosphate bonds
- d. Concentration of nucleotides

Which of the statement given above is/are correct ?

Options :

- 1. c, b and d
- 2. a and c only
- 3. d only
- 4. b only

Question Number : 54 Question Id : 464662854 Question Type : MCQ

Carrier proteins that bind one or more solute molecules on one side of the membrane and then undergo a conformational change to transfer the solute to the other side of the membrane is also called as _____.

Options :

1. Transporter proteins
2. Channel proteins
3. Porins
4. Proton pump

Question Number : 55 Question Id : 464662855 Question Type : MCQ

The oocytes of meroistic insects do not pass through a transcriptionally active stage, nor do they have lamp brush chromosomes, instead RNA synthesis is largely restricted to _____.

Options :

1. the cytocysts
2. the nurse cells
3. the cord cells
4. the stem cells

Question Number : 56 Question Id : 464662856 Question Type : MCQ

Mechanical strength on tissue is conferred by _____.

Options :

1. Microtubules
2. Microfilaments
3. Intermediate filaments
4. Integrins

Question Number : 57 Question Id : 464662857 Question Type : MCQ

A group of cells whose position and fate are specified with respect to the same set of boundaries is described as _____.

Options :

1. Morphogen gradient
2. Syncytial specification
3. Morphogenetic field
4. Histotypic aggregation

Question Number : 58 Question Id : 464662858 Question Type : MCQ

Coat colour inheritance, which is a classic case of recessive epistasis, is seen in which one of the dog breeds given below ?

Options :

1. Pomeranian
2. German Shepherd
3. Golden Retriever
4. Pug

Question Number : 59 Question Id : 464662859 Question Type : MCQ

The start checkpoint in budding yeast mainly determines whether the cell should _____.

- a. enter meiosis
- b. begin cytokinesis
- c. proceed to S phase
- d. enter M phase directly

Which of the statement given above is/are correct ?

Options :

- 1. c only
- 2. a and b only
- 3. b and c only
- 4. a, b and d

Question Number : 60 Question Id : 464662860 Question Type : MCQ

Which statement regarding the CTD (C-terminal domain) of RNA polymerase II is incorrect ?

Options :

1. It consists of repeated heptapeptide sequences.
2. Its phosphorylation regulates transitions during transcription.
3. Its dephosphorylation regulates transitions during transcription.
4. Not directly catalyzes intron excision.

Question Number : 61 Question Id : 464662861 Question Type : MCQ

The morphogenetic furrow movement in Drosophila eye is driven by interaction between _____.

- a. Notch and Wnt
- b. Hedgehog
- c. Decapentaplegic
- d. LIN-3

Which of the statement given above is/are correct ?

Options :

- 1. b and c only
- 2. a and c only
- 3. a and d only
- 4. a and b only

Question Number : 62 Question Id : 464662862 Question Type : MCQ

In the cloning experiment, what is the role of selectable gene markers such as ampR ?

Options :

1. They help in the visualization of colony
2. They cut the DNA at specific sites
3. They join DNA fragments together
4. They allow only cells containing the vector to survive

Question Number : 63 Question Id : 464662863 Question Type : MCQ

A metabolite of which vitamin is responsible for the patterning of proximo-distal axis of the vertebrate limb ?

Options :

1. Vitamin A
2. Vitamin C
3. Vitamin D
4. Vitamin E

Question Number : 64 Question Id : 464662864 Question Type : MCQ

Which of the following enzymes is typically absent in the lysosomes ?

Options :

1. Lipases
2. Proteases
3. Aldolases
4. Sulphatases

Question Number : 65 Question Id : 464662865 Question Type : MCQ

Insertion of a foreign DNA into the BamHI site of the tetracyclin resistance gene. in the plasmid pBR322 results in _____.

Options :

1. Loss of ampicillin and tetracyclin resistance in the recombinant plasmid.
2. Resistance of the recombinant plasmid to both ampicillin and tetracyclin.
3. The recombinant plasmid retaining resistance to ampicillin but sensitive to tetracyclin.
4. The recombinant plasmid becoming sensitive to ampicillin while remaining tetracyclin resistant.

Question Number : 66 Question Id : 464662866 Question Type : MCQ

Which of the following kinases recognises double-strand breaks of DNA during cell cycle ?

- a. Cyclin-dependent protein kinase
- b. ATR kinase
- c. ATM kinase
- d. MDM kinase

Options :

1. Only a
2. Only a and b
3. Only a, b and d
4. Only b and c

Question Number : 67 Question Id : 464662867 Question Type : MCQ

Which combination of histone amino acid residues, when methylated, marks a region of open, transcriptionally active euchromatin ?

Options :

1. H3K27 and H3K20
2. H3K27 and H3K29
3. H3K4 and H3K29
4. H3K4 and H3K36

Question Number : 68 Question Id : 464662868 Question Type : MCQ

Which cellular rearrangement/movement drives the epiboly of ectoderm during amphibian gastrulation ?

Options :

1. Convergent extension
2. Invagination
3. Involution
4. Radial intercalation

Question Number : 69 Question Id : 464662869 Question Type : MCQ

Inhibitory phosphorylation of cdks commonly occurs on which residues ?

- a. Serine 3
- b. Threonine 14
- c. Tyrosine 15
- d. Lysine 48

Which of the statement given above is/are correct ?

Options :

- 1. a, b and d
- 2. b only
- 3. b and c only
- 4. c only

Which of the following statements is incorrect for Robertsonian translocations ?

- a. It involves the reciprocal exchange of segments between the non-homologous long arms of two completely metacentric chromosomes.
- b. It can be clinically diagnosed by preparing a G-banding technique.
- c. It can be clinically diagnosed by preparing a c-banding technique.
- d. Familial down syndrome can arise when an unaffected carrier passes a fused 14/21 chromosome along with normal chromosome 21 to their off spring.

Which of the following combinations is incorrect ?

Options :

- 1. Only a, c and d
- 2. Only b and d
- 3. Only a and c
- 4. Only b, c and d

Question Number : 71 Question Id : 464662871 Question Type : MCQ

Which of the following statements regarding vulval development in C. elegans is correct ?

Options :

1. LIN-3 directly activates the notch signalling pathway
2. LIN-3 secreted by the anchor cell activates the EGFR-Ras/MAPK pathway in vulval precursor cells
3. Notch signalling induce LIN-3 secretion from vulval precursor cells
4. Wnt signalling alone specifies vulval fate

Question Number : 72 Question Id : 464662872 Question Type : MCQ

Which of the following genetic disorders is not caused due to alteration in the number of chromosomes ?

- a. Down's syndrome
- b. Cri-du-chat syndrome
- c. Turner's syndrome
- d. Klinefelter's syndrome

Options :

- 1. only a
- 2. only a, c and d
- 3. only b and c
- 4. only b

Question Number : 73 Question Id : 464662873 Question Type : MCQ

Which of the following signalling molecules acts as a morphogen in specifying the anterior-posterior axis of vertebrate limb ?

Options :

1. Wnts
2. Notch
3. Hedgehog (sonic)
4. FGF

Question Number : 74 Question Id : 464662874 Question Type : MCQ

In animal cells, which cyclin-cdk complex is most important for passage through the restriction point ?

- a. cdk1 – cyclin B
- b. cdk2 – cyclin A
- c. cdk4/6 – cyclin D
- d. cdk2 – cyclin B

Which of the statement given above is/are correct ?

Options :

- 1. a, b and c
- 2. d only
- 3. a and d only
- 4. c only

Question Number : 75 Question Id : 464662875 Question Type : MCQ

Loss of Wnt4 in XX embryos leads to _____.

- a. Normal ovary development
- b. Testis like characteristics
- c. No reproductive organs
- d. Hormonal imbalance

Which of the statement given above is/are correct ?

Options :

- 1. a, c and d
- 2. b and d
- 3. d only
- 4. b only

Question Number : 76 Question Id : 464662876 Question Type : MCQ

Name the test employed to check the purity of butter through the estimation of volatile acids

Options :

1. Ionic number
2. Reichert – Meissl number
3. Saponification number
4. Acid number

Question Number : 77 Question Id : 464662877 Question Type : MCQ

Which of the following statements are correct ?

- i. Endocrine hormones are released into the blood and carried to target cells.
- ii. Eicosanoid is an example of endocrine hormone.
- iii. Paracrine hormones are released into the extracellular space and diffuse to neighbouring target cell.
- iv. Insulin is an example of paracrine hormone.
- v. Autocrine hormones are released by and affect the same cell.

Options :

- 1. i, ii, iii
- 2. i, iii, iv
- 3. i, iii, iv, v
- 4. i, ii, iii, iv, v

Question Number : 78 Question Id : 464662878 Question Type : MCQ

_____ hormones are released into the extracellular space and diffuse to neighbouring target cells.

Options :

1. Apocrine
2. Autocrine
3. Endocrine
4. Paracrine

Question Number : 79 Question Id : 464662879 Question Type : MCQ

Innate (Non-specific) immunity consists of _____ types of barriers.

Options :

1. One
2. Two
3. Three
4. Four

Question Number : 80 Question Id : 464662880 Question Type : MCQ

Cortisol mediates the bodies response to _____ stresses.

Options :

1. Long-term
2. Shorter term
3. Very shorter term
4. None of these

Question Number : 81 Question Id : 464662881 Question Type : MCQ

In nervous system, the centre controlling temperature is situated in the _____ and involves a group of nerve cells in the hypothalamus called heat regulatory centre.

Options :

1. Cerebellum
2. Cerebrum
3. Optic lobes
4. None of these

Question Number : 82 Question Id : 464662882 Question Type : MCQ

About 1200 ml of air is always remain inside the human lungs. It is described as _____.

Options :

1. Tidal volume
2. Residual volume
3. Expiratory reserve volume
4. Inspiratory reserve volume

Question Number : 83 Question Id : 464662883 Question Type : MCQ

Which of the following from a to e, the correct sequence of breathing mechanism of the animals ?

- a. Diffusion of O_2 and CO_2 into alveoli.
- b. O_2 utilized by the cells for catabolic reaction.
- c. Diffusion of O_2 and CO_2 between blood and tissues.
- d. Transport of gases by the blood.
- e. Release of resultant CO_2 .

Options :

- 1. a, b, c, d, e
- 2. e, c, d, a, b
- 3. d, a, b, c, e
- 4. c, d, a, b, e

Question Number : 84 Question Id : 464662884 Question Type : MCQ

Which areas of the human brain are chiefly responsible for managing emotional reactions (such as fear, anger, pleasure) and sexual behaviours ?

Options :

1. Amygdala and medulla oblongata
2. Limbic system and hypothalamus
3. Cerebellum
4. Pituitary gland

Question Number : 85 Question Id : 464662885 Question Type : MCQ

During resting potential stage K^+ ions are _____ times more inside than outside the cell.

Options :

1. 10
2. 20
3. 40
4. 60

Question Number : 86 Question Id : 464662886 Question Type : MCQ

The human kidney can produce a maximal urine concentration of _____.

- a. 1300 mOsm/L
- b. 1400 mOsm/L
- c. 1500 mOsm/L
- d. 1600 mOsm/L

Options :

- 1. Both a and b
- 2. Both c and d
- 3. Only c
- 4. Only d

Question Number : 87 Question Id : 464662887 Question Type : MCQ

Which type of the research design is most appropriate for applying relationship of the cause of the design ?

Options :

1. Experimental design
2. Historical design
3. Exploratory design
4. Descriptive design

Question Number : 88 Question Id : 464662888 Question Type : MCQ

Deontological ethics, primarily associated with Immanuel Kant, focuses on _____.

Options :

1. Consequences of actions
2. Duties and rights
3. Maximizing happiness
4. Personal benefit

Question Number : 89 Question Id : 464662889 Question Type : MCQ

To study the different aspects of the ecofeminism, which of the following is central claim of ecofeminism ?

Options :

1. Study of ecology should be independent of social structure
2. Only woman can solve the environmental calamities
3. Environmental issues are gender neutral
4. Domination of nature is related to domination of woman

Question Number : 90 Question Id : 464662890 Question Type : MCQ

A researcher selects a sampling design that minimizes cost and ensures low sampling error, but finds that the sample systemically excludes a particular subgroup of population. Which characteristic of a good sample design has been primarily violated ?

Options :

1. Viability in terms of available funds
2. Minimizing of sampling error
3. Control of systematic bias
4. Representativeness with generalizability

Question Number : 91 Question Id : 464662891 Question Type : MCQ

The scientists motivation for carrying out an experiment, the experimental design, the results obtained and the meaning, interpretation and significance of the results is known as _____.

Options :

1. Research plan
2. Research report
3. Research format
4. None of these

Question Number : 92 Question Id : 464662892 Question Type : MCQ

If a series of data in an array is divided into 10 equal parts with nine dividing points, each dividing point is called as _____.

Options :

1. Monile
2. Quadrile
3. Decile
4. Mean

Question Number : 93 Question Id : 464662893 Question Type : MCQ

_____ is a sample drawn in such a way as to insure that every number of the population has an equal chance of being included in the sample.

Options :

1. Cluster-sampling
2. Random sampling
3. Stratified sampling
4. Quota sampling

Question Number : 94 Question Id : 464662894 Question Type : MCQ

The level of significance is justified by which statement from given below ?

Options :

1. Probability to rejecting a true null hypothesis
2. Probability of failing to reject false null hypothesis
3. Probability of accepting true null hypothesis
4. Probability of rejecting false null hypothesis

Question Number : 95 Question Id : 464662895 Question Type : MCQ

Rejecting a true null hypothesis is called _____.

Options :

1. Type II error
2. Type I error
3. Sampling error
4. Standard error

Question Number : 96 Question Id : 464662896 Question Type : MCQ

Database Tr EMBL is providing data of protein sequences obtained by translation of all coding sequences in _____ database.

Options :

1. Australian molecular biology laboratory
2. Japanese molecular biology laboratory
3. European molecular biology laboratory
4. None of these

Question Number : 97 Question Id : 464662897 Question Type : MCQ

In bioinformatics KEGG is the primarily resource of which of the following ?

Options :

1. Protein structure database
2. Phylogenetic database
3. Sequence of nucleotide repository
4. Metabolic pathway and systems biology database

Question Number : 98 Question Id : 464662898 Question Type : MCQ

In many cases, cDNA sequences are more informative than gene sequences because _____.

Options :

1. The RNA molecule may be extensively processed during synthesis
2. Introns may be spliced out of a primary transcript to generate mRNA
3. Extra nucleotide may be inserted by RNA editing
4. All the above

Question Number : 99 Question Id : 464662899 Question Type : MCQ

_____ software is used as bioinformatic tool used for analysis and implementation of genome sequence data for production of function of a new gene.

Options :

1. FASTA
2. SMART
3. GAME
4. DAS

Question Number : 100 Question Id : 464662900 Question Type : MCQ

_____ is a collection of confirmed protein sequences with annotation related to structure, function and protein family assignments.

Options :

1. Swiss-prot
2. Tr EMBL
3. Uni-prot
4. Prosite